DOCTORAL APPLICATION TECHNICAL PORTFOLIO

From Calibrated Forecasts to Risk-averse Storage Decisions

A reproducible mean-CVaR benchmark for day-ahead battery scheduling

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SIMULATED BENCHMARK - NOT A TRADING SYSTEM. All prices, profits, losses and regret values are simulated. They are not employer evidence or realised market revenue.

Abstract

Project 1 showed that a forecast interval can look better statistically after calibration. This project asks whether that change leads to a better battery decision. It uses the 83-day simulated evaluation block from Project 1 and a linear battery model with state of charge (SOC), efficiency, power limits and required end-of-day SOC. Seven schedules are compared, including risk-neutral and CVaR versions. Adaptive-CVaR earns 9.86 simulated EUR/day, compared with 9.73 for raw-CVaR and 8.20 for static-CVaR. Its empirical 90%-CVaR loss is 16.23 EUR, with no violations of the simplified constraints. A wider interval is not automatically a better input for a decision. The benchmark uses fixed-probability-mass CVaR. This documentation revision clarifies scenario scaling and verification history; numerical results are unchanged.

1. Motivation and contribution

At Tarim Oilfield I saw forecast curves, actual photovoltaic output and storage monitoring screens. I did not design the storage schedule, but the work made one point clear: a charging action now changes the energy available later. This experience is only the motivation for the project. No field data are used. I therefore focus on errors near an SOC or power boundary. The same price error can be harmless when the battery has room, but costly when it changes a charge or discharge decision.

Two choices keep the comparison inspectable. First, the linear battery model restores terminal SOC to initial SOC, preventing profit from borrowing end-of-day energy value. Second, raw, static and adaptive forecast paths use the same battery and scoring rules. The contribution is an auditable link from forecast uncertainty to constrained decisions, not a new CVaR or bidding formulation.

2. Decision model

2.1 Battery physics

The battery can charge or discharge at up to 1 MW and can store 2 MWh. SOC increases by the energy that survives charging losses and falls by the energy needed for discharge. Both efficiencies are 0.94. SOC must stay between zero and 2 MWh. It starts and ends at 1 MWh. A 2 EUR/MWh throughput cost makes unnecessary cycling less attractive.

$SOC_{t+1} = SOC_t + 0.94\ charge_t - discharge_t / 0.94$, with $0 <= SOC <= 2$ and $0 <= charge, discharge <= 1$.

2.2 Scenario profit, optimisation and CVaR reporting

Profit is discharge revenue minus charging and throughput costs. The risk-neutral objective maximises average scenario profit; the risk-aware objective subtracts weight times loss CVaR, with alpha 0.90 and base weight 0.55. The Rockafellar-Uryasev auxiliary-variable formulation is solved with SciPy/HiGHS: CVaR = min over zeta of zeta + $sum(u_s)/((1 - alpha)\ N)$, with $u_s >= loss_s$ - zeta and $u_s >= 0$.

Reporting definition introduced in v1.0.1. Sort losses from largest to smallest. Let m = (1 - alpha) n, k = floor(m) and r = m - k. CVaR is (sum of the k largest losses + r times the next loss)/m; omit the next loss when r is zero. Thus 60 equally weighted scenarios contribute six observations and 83 evaluation days contribute 8.3 observations. This preserves fixed tail probability mass with ties and fractional boundaries (Rockafellar and Uryasev, 2002). The LP optimisation formula is unchanged.

2.3 Feasibility and assumptions

All seven strategies over 83 days have zero reported constraint violations and zero simultaneous charge/discharge throughput. Tests independently recompute SOC transitions and terminal SOC. These checks apply to the simplified benchmark constraints.

Assumption	Reason	Consequence
Price taker	Keeps the first portfolio auditable	No market-impact or bid-acceptance model
Single energy market	24-hour scheduling horizon	No intraday or balancing recourse
Hourly perfect execution	Separates forecast value from controls	No deviation settlement or outage
Linear throughput cost	Transparent degradation proxy	No cycle-depth or ageing state

2.4 Relationship to existing work

CVaR optimisation is established (Rockafellar and Uryasev, 2000, 2002). Donti et al. (2017) and Elmachtoub and Grigas (2022) connect prediction and decision loss. Kim et al. (2021) and Toubeau et al. (2021) study uncertain storage scheduling; Yeh et al. (2025) and Alghumayjan et al. (2025) connect conformal uncertainty to decisions. This portfolio integrates those ideas with profit, regret, tail risk and feasibility checks.

3. Forecast-to-scenario and experimental design

Project 1 supplies seven marginal quantiles per hour. Interpolation produces 60 possible 24-hour paths. A shared Gaussian factor has loading 0.65, implying latent Pearson correlation 0.4225 (Spearman about 0.41) before clipping and interpolation. Dependence is exchangeable rather than lag-specific. Static and adaptive quantile curves are rescaled around the median using scale = clip(calibrated width / max(q90 - q10, 1e-6), 0.35, 4.0). Their q10-q90 widths match the calibrated interval only when the width floor and scale clipping are inactive. Of 1,992 hourly rows, four static and seven adaptive scales reach the upper cap; none reaches the lower cap. A finite sample of 60 paths need not reproduce these quantile widths exactly. This construction does not establish joint calibration of 24-hour paths.

Strategy	Forecast information	Objective
Seasonal naive	Previous-day price	Risk neutral
Median risk neutral	q50	Risk neutral
Raw CVaR	Raw quantile paths	Mean-CVaR
Static CVaR	Static-CQR-scaled paths	Mean-CVaR
Adaptive risk neutral	Adaptive-scaled paths	Risk neutral
Adaptive CVaR	Adaptive-scaled paths	Mean-CVaR
Oracle	Realised prices	Risk neutral upper bound

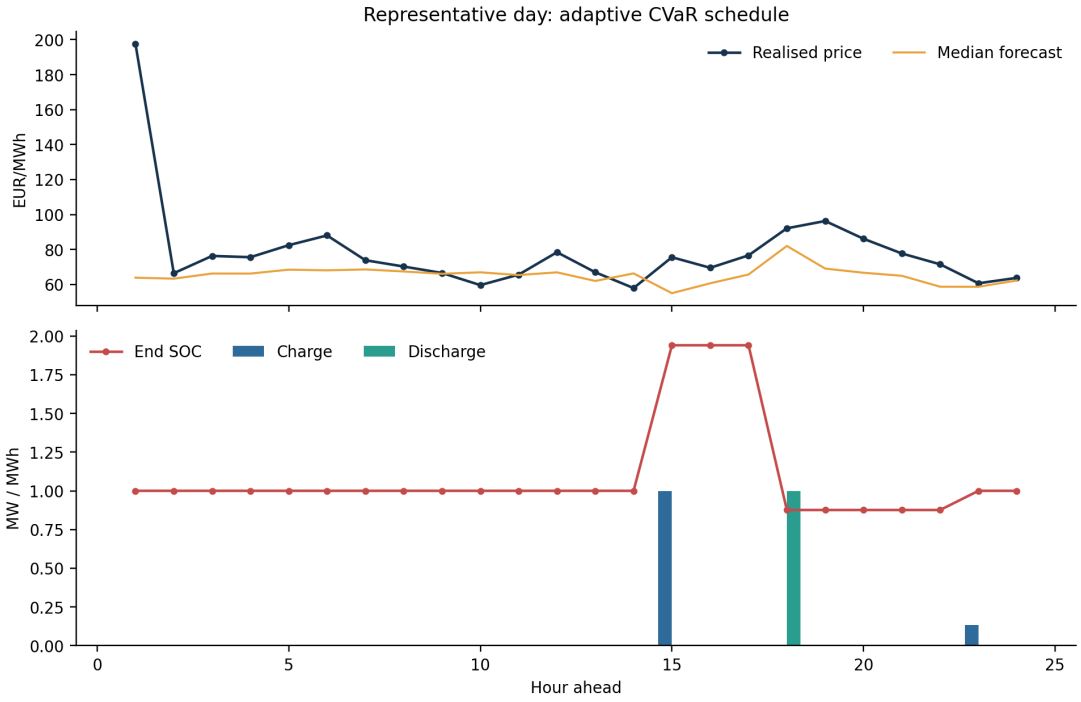


Figure 1. Representative adaptive-CVaR schedule, showing power and SOC feasibility and restoration of terminal SOC. Source: corrected v1.0.1 baseline, retained in v1.0.2.

The chosen schedule uses forecast information; realised prices are used to score it. The oracle is a deliberately infeasible information benchmark. Operational issuance-time limitations are stated in Section 7.

4. Main results

All financial values below are simulated EUR. The main comparison uses 83 evaluation days at CVaR weight 0.55. CVaR uses the worst 10% loss probability mass. Mean profit and regret are per day.

Strategy	Mean profit	5th pct profit	CVaR loss	Mean regret
Seasonal naive	-28.28	-80.73	82.64	84.93
Median risk neutral	8.34	-18.33	23.14	48.31
Raw CVaR	9.73	-19.79	21.90	46.91
Static CVaR	8.20	-17.17	20.42	48.45
Adaptive risk neutral	9.18	-19.23	23.18	47.46
Adaptive CVaR	9.86	-15.24	16.23	46.79
Oracle	56.65	26.24	-24.68	0.00

The seasonal rule loses 28.28 EUR/day. The median forecast produces positive mean profit. Raw-CVaR earns 9.73 with CVaR loss 21.90. Static-CVaR lowers tail loss to 20.42 but earns 8.20. Adaptive-CVaR combines 9.86 mean profit with CVaR loss 16.23, the lowest tail loss among non-oracle strategies in this base comparison.

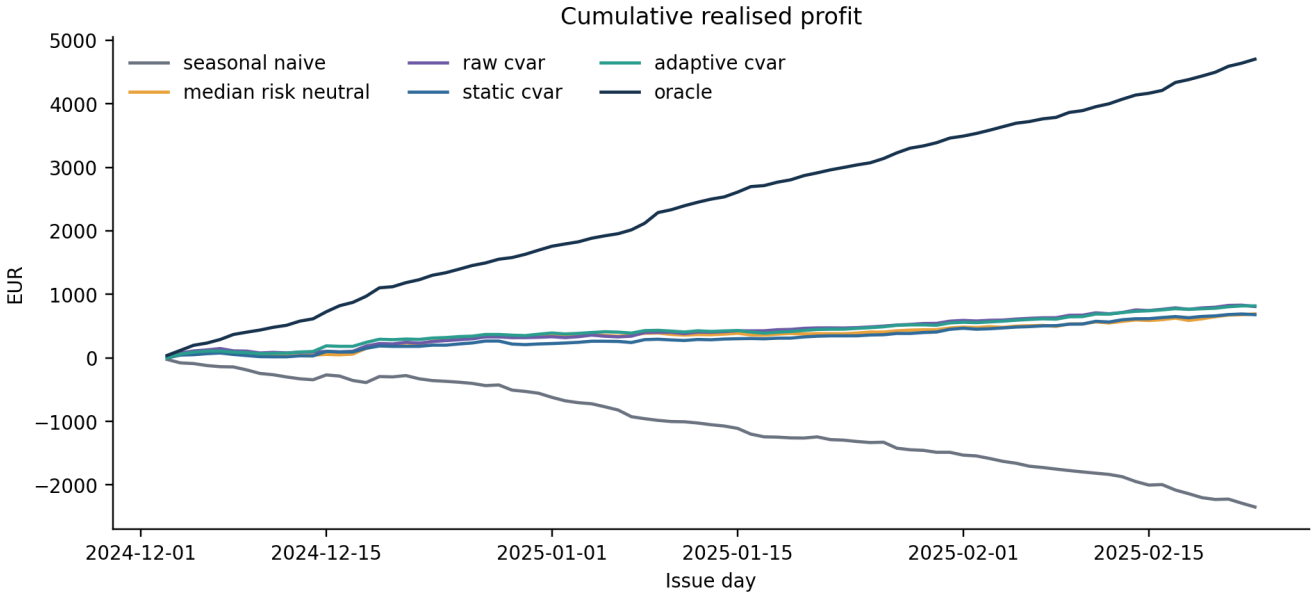


Figure 2. Cumulative simulated profit over the evaluation block; profit is unchanged by the reporting correction. Source: corrected v1.0.1 baseline, retained in v1.0.2.

The oracle earns 56.65 EUR/day, leaving substantial regret for forecast-based strategies. It requires future realised prices and is not a realistic trading opportunity. Negative loss CVaR means even its selected worst tail remains profitable in this simulation.

5. Tail risk and regret

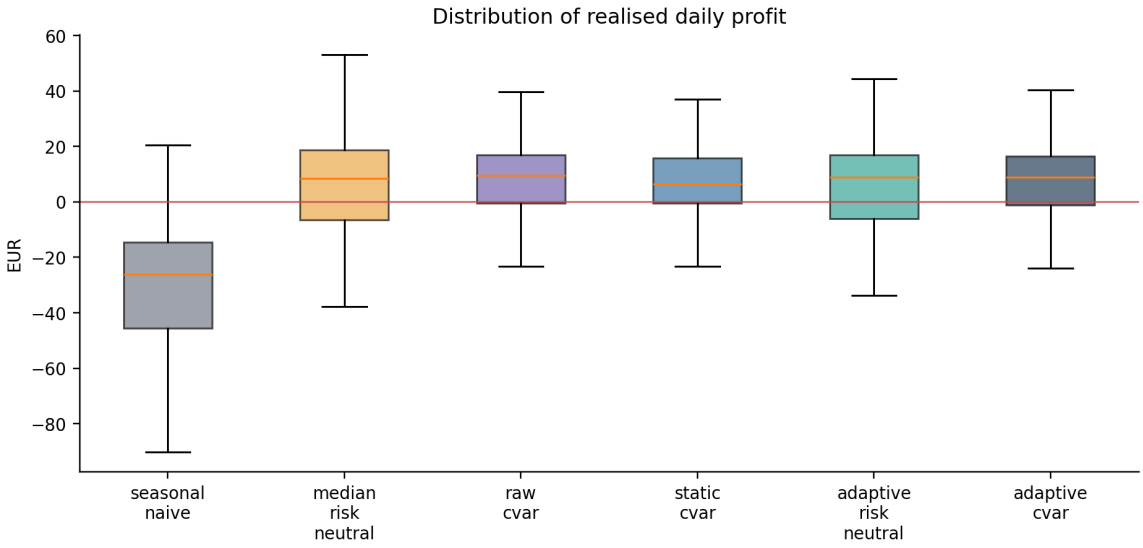


Figure 3. Simulated daily profit distributions for feasible benchmark strategies. Source: corrected v1.0.1 baseline, retained in v1.0.2.

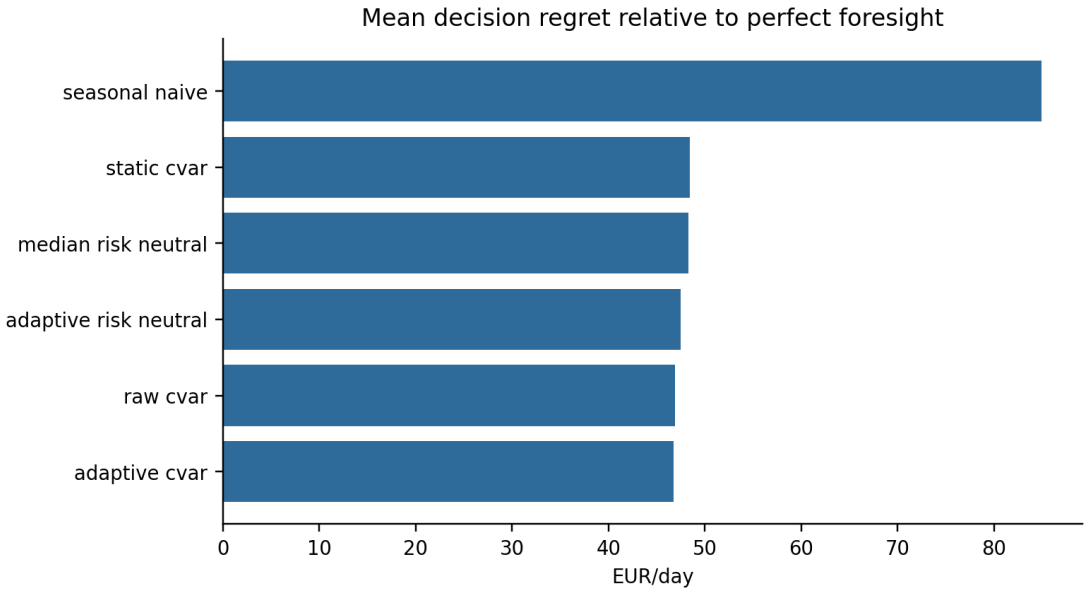


Figure 4. Mean daily regret relative to the perfect-foresight oracle; lower is better. Source: corrected v1.0.1 baseline, retained in v1.0.2.

5.1 What calibration changed

Static calibrated paths reduce tail losses but give up some profit relative to raw-CVaR. Adaptive-CVaR gives a more favourable balance in this base simulation. Coverage alone is not a decision metric: calibration changes path dispersion, which changes dispatch under efficiency, throughput cost and SOC constraints. These results do not establish the same ranking in another market or seed.

5.2 CVaR-weight sensitivity

Alongside the supplied 0.55 and 0.85 settings, I ran a pre-planned 0.70 test on the same simulated benchmark and seed on 2 October. This table uses the corrected reporting definition; original run records retain their earlier diagnostic values.

Weight	Adaptive mean profit	5th pct profit	CVaR loss	Throughput MWh/day
0.55	9.86	-15.24	16.23	4.20
0.70	9.78	-14.87	15.03	4.00
0.85	9.99	-11.09	12.59	3.83

Tail loss and fifth-percentile profit improve across these adaptive-CVaR runs while throughput decreases. Mean profit is non-monotonic. Weight 0.70 reduces tail risk relative to 0.55 with slightly less profit. This single-seed comparison does not establish an optimum; the base remains 0.55.

6. Risk preference and battery parameters

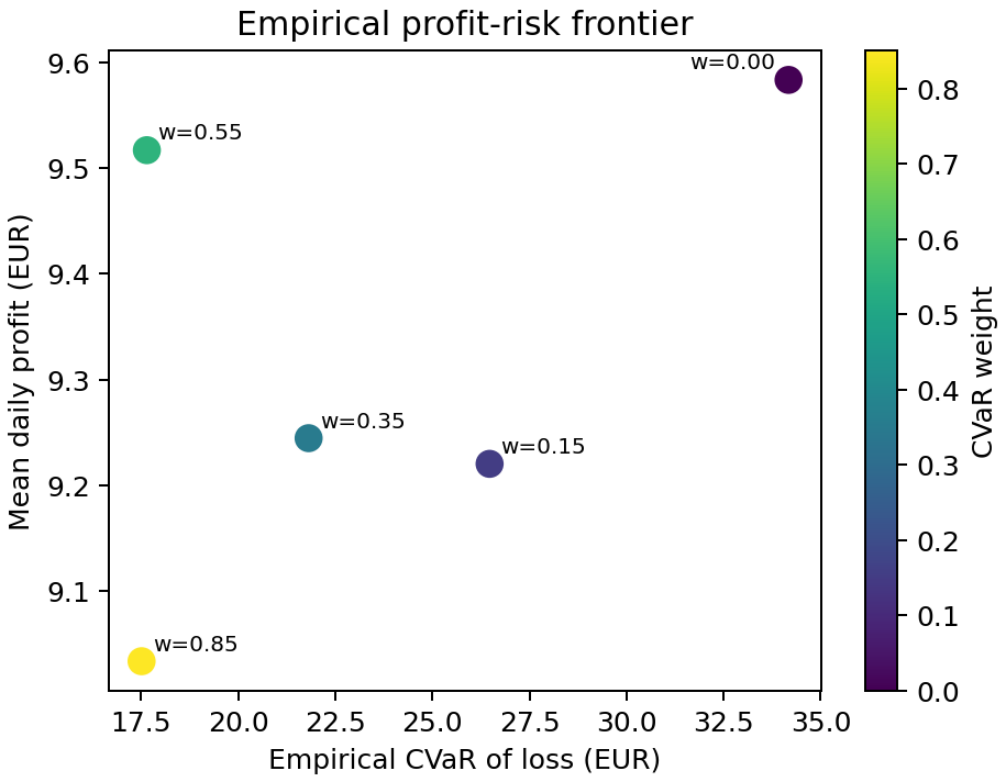


Figure 5. Mean simulated profit versus fixed-mass empirical CVaR loss, using a separate frontier scenario sample. Source: corrected v1.0.1 baseline table; labels redrawn for v1.0.2.

In the frontier sample, increasing the weight from 0 to 0.85 reduces CVaR loss from 34.17 to 17.51 EUR. Mean profit at 0.85 is 9.03, compared with 9.52 at 0.55. These values differ from the main sensitivity runs because the frontier draws its own scenario sample. Non-monotone profit reflects finite scenarios and one realised evaluation path.

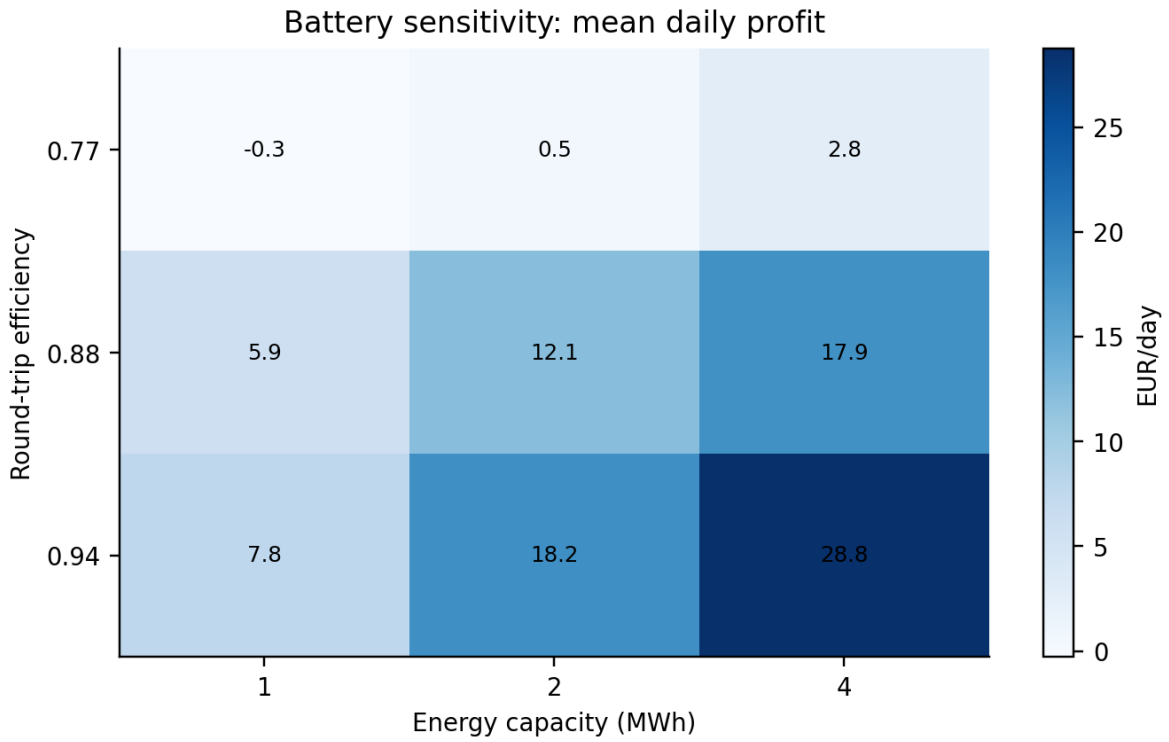


Figure 6. A 28-day diagnostic of median-forecast profit across energy capacity and efficiency. Source: corrected v1.0.1 baseline, retained in v1.0.2.

Higher efficiency consistently improves value; added capacity helps most when efficiency is high. This is diagnostic, not an investment valuation, because capex and replacement costs are excluded.

7. What the project still cannot answer

7.1 Limitations and timing boundary

The biggest limit is the data: prices and forecasts are simulated. The second is the scenario model. Calibrating each hour does not guarantee that an entire 24-hour path is realistic. The third is the battery model. It assumes price-taking, perfect execution and no intraday update. It also leaves out network limits, imbalance settlement and detailed degradation. Finally, CVaR is estimated from only 60 scenarios. Different seeds, correlations or markets may change the ranking.

Issuance-time boundary. In the supplied CSV, issue_time equals target_time for horizon 1. A separate prior-day market gate-closure timestamp is not recorded. This 24-hour scheduling benchmark therefore does not itself verify operational day-ahead forecast availability. A real-data study would need publication-time checks before any bidding claim.

The 83-day block was held out from Project 1 model fitting and conformal calibration, but Project 1 also used it for a post-hoc gamma sensitivity diagnostic; it is therefore not an untouched test period.

7.2 Reproducibility and revision record

The manifest records seed 20260817, 83 days, 60 scenarios per day, battery settings, software and the estimator identifier. Current snapshots are in experiments/baseline, experiments/cvar_weight_070 and experiments/cvar_weight_085. The numerical snapshots were generated for v1.0.1 and are retained unchanged. Figure 5 is redrawn from the same table to improve label placement; the other figures retain their saved pixels.

```
python -m unittest discover -s tests -v
python scripts/verify_release.py
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Verification history. The Linux/Python 3.12.14 review on 4 October passed 11 unit tests and 9/9 table comparisons at absolute tolerance 1e-10. Its 3,744 paired old/new diagnostic solves had identical schedules, expected profits and LP objectives. The applicant subsequently passed the same 11 tests and nine comparisons on Windows/Python 3.13.15. The recorded CI run for commit c6155e7 passed all six Ubuntu/Windows jobs with Python 3.11, 3.12 and 3.13; its status was recorded on 5 October. These are historical checks of the corrected numerical implementation, not a CI claim for a later commit.

On 9 October, a fresh Linux/Python 3.12.14 review passed all 11 tests and reproduced all 15 saved CSV tables across the three weights; all 18 regenerated figures matched the saved pixels. This v1.0.2 revision retains the numerical implementation and reference outputs. Personal logs, review records and exact CI links remain indexed in REFERENCE_RUNS.md.

7.3 Doctoral extension

My next step would be to repeat the study on publication-time-safe market data and more than one seed. I would then improve the 24-hour scenario model and add intraday recourse and market-specific settlement. The main doctoral question stays the same: when does better uncertainty information lead to a better constrained decision, especially near SOC boundaries?

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